target. For any re-entry missile system, three-dimensional preforms are core material for making the carbon-carbon nose tip that wil remain strong even at such high temperatures. Four laboratories of DRDO and the CSIR achieved this in a short span of 18 months— something other countries could do only after a decade of research and development!

Another chal enge involved in the Agni payload design was related to the tremendous speed with which it would re-enter the atmosphere. In fact, Agni would re-enter the atmosphere at twelve times the speed of sound (12 Mach, as we cal it in science). At this tremendous speed, we had no experience of how to keep the vehicle under control. To carry out a test, we had no wind tunnel to generate that kind of speed. If we sought American help, we would have been seen as aspiring to something they considered their exclusive privilege. Even if they consented to co-operate, they would be certain to quote a price for their wind tunnel greater than our entire project budget. Now, the question was how to beat the system. Prof. SM Deshpande of the I Sc found four young, bright scientists working in the field of fluid dynamics and, within six months, developed the